

SkinDisNet-GT: Graph-Transformer-Enhanced Skin Disease Classification from Clinical Images

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Abstract—Skin diseases are common and harm quality of life, yet diagnosis depends on scarce dermatological expertise. Automated classification of clinical images is hindered by variable illumination, background, acquisition, and appearance, and by class imbalance. We propose SkinDisNet-GT, a graph-transformer framework combining hierarchical visual features with explicit spatial modeling, evaluated on the SkinDisNet clinical dataset of 1,710 images from six classes. A pretrained Swin-Tiny backbone maps 224×224 inputs to a $14 \times 14 \times 384$ feature map, converted into 196 nodes of a nine-nearest-neighbor graph processed with degree-based structural encoding, spatial-distance bias, multi-head graph attention, and graph-transformer blocks. Class-balanced sampling, weighted focal loss, label smoothing, Mixup/CutMix, two-stage fine-tuning, and test-time augmentation address imbalance. On the held-out test set, SkinDisNet-GT achieved 87.94% accuracy, 0.8761 weighted F1, 0.9534 weighted AUC-ROC, and 0.6060 test loss, with Cohen’s κ of 0.8443 and MCC of 0.8484. In controlled ablations, the advanced-training Swin-Tiny configuration reached 89.11%. The study shows the viability of hierarchical transformer features for clinical skin-disease diagnosis, with quantitative and visual interpretability analysis.

Index Terms—Skin disease classification, clinical dermatology, deep learning, graph transformer, Swin Transformer, graph neural network, attention mechanism, SkinDisNet, medical image analysis.

I. INTRODUCTION

Skin diseases affect people of any age, nationality, and skin type. Diagnosis relies on experienced doctors using clinical knowledge, visual examination, dermatoscopy, and laboratory tests, but timely access to a specialist is not always possible, motivating automated systems that identify the disease from a photograph. Deep learning has shown great promise here, from the HAM10000 benchmark [1] and clinical-image skin cancer studies [2] to multi-class CNNs under imbalance [3] and segmentation-based pipelines [4]. Vision Transformers [5] and hierarchical, contextual, and hybrid transformer-CNN models have been studied in dermatology [7]–[9], while graph-based vision treats image regions as nodes [6], [11] and graph attention and graph transformers add relational information and structural biases [12], [13].

This work explores explicitly organizing hierarchical transformer features as a graph for clinical skin disease classification on SkinDisNet, a six-class clinical dataset. Unlike controlled dermoscopic datasets, clinical photographs vary in illumination, background, and appearance, and the dataset is highly imbalanced, so balanced learning and augmentation are required.

We aim to (i) develop SkinDisNet-GT; (ii) build an imbalance-aware pipeline with class-balanced sampling, weighted focal loss, label smoothing, Mixup/CutMix, and two-phase fine-tuning; (iii) evaluate accuracy, precision, recall, weighted F1, one-vs-rest AUC, test loss, confusion matrices, and per-class results; (iv) perform controlled ablations of graph reasoning, Mixup/CutMix, and test-time augmentation (TTA); (v) provide Grad-CAM visualizations of the regions relevant to predictions; and (vi) analyze data-splitting and augmentation leakage controls and identify remaining limitations in patient-level grouping.

Our contributions are (1) a Swin-Tiny-to-graph architecture mapping a 14×14 hierarchical feature grid to 196 relational image-region nodes; (2) a graph-transformer module integrating KNN aggregation, degree-based structural encoding, spatial-distance bias, multi-head attention, and feed-forward transformation; (3) a controlled study of advanced training components and graph removal on one evaluation partition; and (4) a Grad-CAM analysis of the graph-transformer features.

II. RELATED WORK

CNN-based classification. HAM10000 is a multi-source dermatoscopic benchmark of 10,015 images in seven categories [1]; end-to-end deep networks succeeded at clinical skin cancer classification [2]; ResNeXt-101 performed well among CNNs on seven-class HAM10000 [3]; and graph-cut segmentation with Naïve Bayes gave an efficient but handcrafted pipeline [4]. These methods miss relational modeling: CNNs are local, and handcrafted pipelines separate lesion isolation from recognition.

Transformers in dermatology. After ViT [5], DermViT used contextual and hierarchical attention for lesion variability and localization [8], Swin Transformers performed well on mobile-captured images under uncontrolled acquisition [7], and EViT-Dens169 fused transformer and CNN features with hybrid attention [9]. Our baselines follow this trend: ViT-B/16 reached 79.77% accuracy (0.7948 weighted F1) and Swin-Tiny 84.44% (0.8406), a direct reference for the benefit of graph reasoning.

Graph-based vision. Vision GNN builds nearest-neighbor graphs over patches [6]; Progressive Vision Graph addresses first-order similarity, aggregation cost, and deep-graph behavior [11]; GAT learns neighbor attention instead of fixed

TABLE I
SKINDISNET CLASS DISTRIBUTION

Class	N	Share	Class	N	Share
Atopic Derm.	70	4.09%	Scabies	343	20.06%
Contact Derm.	477	27.89%	Seborrheic Derm.	79	4.62%
Eczema	466	27.25%	Tinea Corporis	275	16.08%
Total: 1,710 images			416 patients; 1–37 (mean 4.11) each		

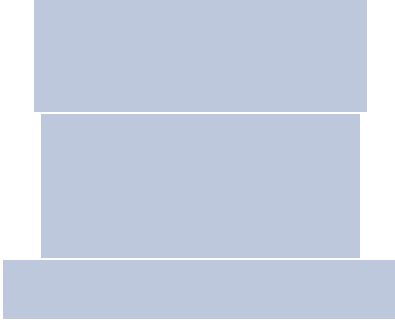


Fig. 1. Sample images from the six skin disease classes.

weights [12]; Graphormer adds structural and positional features to attention [13]; and a review identifies graph construction and graph transformers as key parts of medical image pipelines [14]. SkinDisNet-GT places the graph module after, not instead of, a hierarchical Swin backbone.

Multimodal work. VitAtt fuses image tokens with patient metadata through interpretable attention [10]. SkinDisNet also has metadata, but no metadata CSV was found alongside the image folders in the baseline notebook, so all results here are image-only. The gap is that graph-transformer processing is understudied for realistic clinical skin images such as SkinDisNet.

III. PROPOSED METHODOLOGY: SKINDISNET-GT

A. Dataset and Class Distribution

SkinDisNet contains 1,710 preprocessed images from 416 patients with patient metadata, covering six classes: Atopic Dermatitis (AD), Contact Dermatitis (CD), Eczema (EC), Scabies (SC), Seborrheic Dermatitis (SD), and Tinea Corporis (TC), with 70, 477, 466, 343, 79, and 275 images (Table I, Figs. 1 and 2a). CD and EC dominate while AD and SD are minorities. Each patient has 1 to 37 images (mean 4.11), so several images may show different lesion regions, viewing angles, or clinical situations of the same patient (Fig. 2b); grayscale intensity distributions per class are shown in Fig. 2c.

A stratified 70:15:15 split gave 1,197 training, 256 validation, and 257 test images; offline augmentation of the training set only gave 9,576 training samples. Base-image identification kept each original and its augmented copies in one partition, so partitions were non-overlapping and no augmented copy of a validation or test image entered training. Validation and testing used only original preprocessed images. The patient structure was considered during splitting, but strict patient-level separation is not guaranteed (Section VI). Fig. 3 gives an overview of the method.

Input Processing and Augmentation. Inputs are 224×224 RGB with ImageNet normalization (mean $[0.485, 0.456, 0.406]$, standard deviation $[0.229, 0.224, 0.225]$). The SkinDisNet-GT training transform resizes to 256×256 , crops to 224×224 , applies horizontal (0.5) and vertical (0.3) flips, rotation up to 30° , color jitter (brightness 0.3, contrast 0.3, saturation 0.2, hue 0.1), and grayscale with probability 0.05. Baselines use only resizing and horizontal flips. Each batch gets Mixup ($\alpha = 0.2$) or CutMix ($\alpha = 1.0$), equally likely, with probability 0.5. TTA averages predictions on the original and horizontally flipped image.

Swin Feature Extraction. The pretrained timm Swin-Tiny model `swin_tiny_patch4_window7_224`, with feature extraction at output stage index 2, produces a $14 \times 14 \times 384$ grid, flattened into $N = 196$ nodes. Each 384-dimensional node is projected to 256 dimensions by a linear layer and GELU, a learned 256-dimensional positional embedding is added, and LayerNorm follows, giving $X \in \mathbb{R}^{B \times 196 \times 256}$.

K-Nearest-Neighbor Graph Construction. Each image becomes a 196-node graph: after L2 normalization, each node keeps the $k = 9$ most similar other nodes, excluding itself:

$$s_{ij} = \frac{x_i^T x_j}{\|x_i\|_2 \|x_j\|_2}, \quad \mathcal{N}_k(i) = \text{TopK}_{j \neq i}(s_{ij}), \quad (1)$$

with $A_{ij} = 1$ if $j \in \mathcal{N}_k(i)$ and 0 otherwise, $A \in \{0, 1\}^{B \times 196 \times 196}$. This gives an explicit local relational structure while retaining all spatial nodes.

Structural Encoding. Each node is augmented with a learned degree/centrality embedding and a spatial-distance bias. The degree $d_i = \sum_j A_{ij}$ is clipped to 20 and embedded through a learned table, giving $\hat{x}_i = x_i + e_{cent}(d_i)$. Node positions are the original 14×14 grid coordinates, and the integer Euclidean distance between two locations is mapped through a learned scalar embedding to give the attention bias b_{ij} .

Graph Aggregation Module. Each graph-transformer block contains a Grapher module. Node features pass through a linear layer and LayerNorm, and neighbor features gathered by the KNN indices are averaged, $\bar{x}_i = \frac{1}{k} \sum_{j \in \mathcal{N}_k(i)} x_j$. The center and aggregated features are concatenated and passed through a two-layer transformation g with GELU and dropout, with a residual connection:

$$x'_i = x_i + g([x_i; \bar{x}_i]). \quad (2)$$

Graph Transformer Attention. The attention layer has eight heads; queries, keys, and values come from a single linear projection of the normalized node features. For head h ,

$$S_{ij}^{(h)} = \frac{Q_i^{(h)} K_j^{(h)T}}{\sqrt{d_h}} + b_{ij}, \quad \alpha_{ij}^{(h)} = \text{softmax}_j(S_{ij}^{(h)}), \quad (3)$$

$$Z_i^{(h)} = \sum_j \alpha_{ij}^{(h)} V_j^{(h)}. \quad (4)$$

Head outputs are concatenated, projected, and added residually; the spatial bias lets attention use structural position as well as content.

Feed-Forward Transformation and Classification. Each block ends with a residual feed-forward network (LayerNorm,

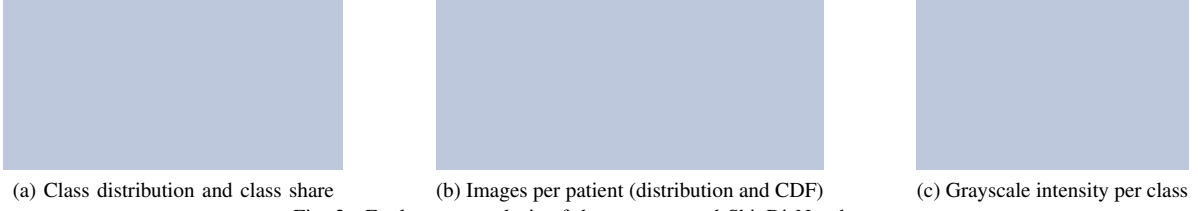


Fig. 2. Exploratory analysis of the preprocessed SkinDisNet dataset.

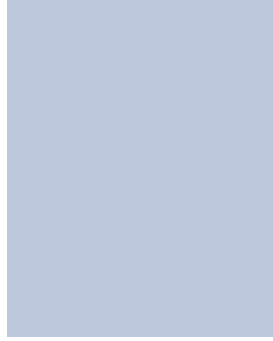


Fig. 3. Overview of the proposed SkinDisNet-GT methodology.

Linear–GELU–Dropout, Linear–Dropout) of width $4 \times 256 = 1024$. Three identical graph-transformer blocks are stacked, followed by LayerNorm and mean pooling over the 196 nodes. The classifier is a two-layer MLP $256 \rightarrow 128 \rightarrow 6$ with GELU and dropout. The model has 16,091,749 parameters, all trainable once the backbone is unfrozen.

Imbalance-Aware Objective. Class weights computed from the training labels are $[4.0714, 0.5973, 0.6120, 0.8313, 3.6273, 1.0337]$ for AD, CD, EC, SC, SD, and TC. The hard-label focal loss applies focusing factor $\gamma = 2$ to weighted cross-entropy with label smoothing 0.1. With CE_i the weighted smoothed cross-entropy of sample i and $p_{t,i} = \exp(-CE_i)$,

$$L_{focal} = \frac{1}{B} \sum_i (1 - p_{t,i})^\gamma CE_i. \quad (5)$$

For Mixup/CutMix batches with soft targets $y_i \in \mathbb{R}^6$ and $p_{ic} = \text{softmax}(z_i)_c$,

$$L_{soft} = \frac{1}{B} \sum_i (1 - p_{t,i})^\gamma \left[- \sum_c y_{ic} \log p_{ic} \right]. \quad (6)$$

The evaluation criterion is weighted cross-entropy without focal modulation.

Optimization and Training Schedule. SkinDisNet-GT is trained for 60 epochs with batch size 16 using AdamW (weight decay 10^{-2}), with learning rates 10^{-5} for the Swin backbone and 3×10^{-4} for all other parameters, and cosine annealing toward 10^{-6} . The backbone is frozen during epochs 1–10 and fully unfrozen from epoch 11. Mixed precision is enabled and gradients are clipped to norm 1.0. The best checkpoint is selected by validation weighted F1, with a best value of 0.9251 at epoch 47. The Swin-Tiny backbone alone has about 28.3M parameters.

IV. EXPERIMENTAL SETUP

A. Baseline Implementations

Three pretrained baselines, ViT-B/16, Swin-Tiny, and EfficientNet-B0, are fine-tuned for six classes under the same 70/15/15 stratified split, 224×224 images, batch size 32, class-weighted cross-entropy, AdamW (learning rate 10^{-4} , weight decay 10^{-2}), cosine annealing to 10^{-6} , mixed precision, and validation-accuracy checkpoint selection, for 30 epochs each. ViT-B/16 and EfficientNet-B0 have 85,803,270 and 4,015,234 parameters, respectively.

B. Evaluation Metrics

Test-set metrics are accuracy, weighted F1, weighted one-vs-rest ROC-AUC, and per-class precision, recall, F1, and support. With TTA, SkinDisNet-GT obtained 87.55% accuracy, weighted F1 of 0.8734, and weighted ROC-AUC of 0.9392. Dice and IoU are not used, since they are segmentation metrics and this study is classification. With predictions $\hat{y}_i$ and labels y_i , $Acc = \frac{1}{N} \sum_i \mathbf{1}[\hat{y}_i = y_i]$ and

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}, \quad F1_c = \frac{2P_c R_c}{P_c + R_c}. \quad (7)$$

The weighted aggregate is $F1 = \sum_c w_c F1_c$, with w_c the class-support fraction, and the macro aggregate is the unweighted mean over the six classes. ROC-AUC uses one-vs-rest probabilities with weighted averaging. We additionally derive one-vs-rest specificity, negative predictive value, balanced accuracy, F_β ($\beta = 2$ weights recall), and the Jaccard index:

$$Sp_c = \frac{TN_c}{TN_c + FP_c}, \quad NPV_c = \frac{TN_c}{TN_c + FN_c}, \quad BA_c = \frac{R_c + Sp_c}{2}, \quad (8)$$

$$F_{\beta,c} = \frac{(1+\beta^2)P_c R_c}{\beta^2 P_c + R_c}, \quad J_c = \frac{TP_c}{TP_c + FP_c + FN_c}. \quad (9)$$

Per-class MCC applies the binary MCC one-vs-rest; multiclass MCC and Cohen's kappa are

$$MCC = \frac{C N - \sum_c \hat{n}_c n_c}{\sqrt{(N^2 - \sum_c \hat{n}_c^2)(N^2 - \sum_c n_c^2)}}, \quad \kappa = \frac{p_o - p_e}{1 - p_e}, \quad (10)$$

TABLE II
OVERALL TEST METRICS OF SKINDISNET-GT (NON-TTA, $N = 257$);
MACRO, κ , MCC, AND CI DERIVED FROM TABLE III

Metric	Value	Metric	Value
Test loss	0.6060	Correct / total	226 / 257
Accuracy (micro-F1)	87.94%	95% Wilson CI	[83.39, 91.37]%
Error rate	12.06%	Weighted AUC-ROC	0.9534
Prec. W/M	.8918/.9035	Cohen's κ (p_e)	0.8443 (0.2255)
Rec. W/M	.8794/.8235	Multiclass MCC	0.8484
F1 W/M	.8761/.8485	Parameters	16,091,749

where C is the number of correct predictions, n_c and $\hat{n}_c$ are the true and predicted counts of class c , p_o is the accuracy, and $p_e = \sum_c n_c \hat{n}_c / N^2$. Accuracy $\hat{p}$ on N images has the 95% Wilson interval ($z = 1.96$)

$$(\hat{p} + \frac{z^2}{2N} \pm z\sqrt{\hat{p}(1-\hat{p})/N + z^2/(4N^2)}) / (1 + z^2/N). \quad (11)$$

V. RESULTS AND ABLATION STUDY

A. Final SkinDisNet-GT Performance

The non-TTA configuration is selected for its better accuracy and weighted F1: test loss 0.6060, 87.94% accuracy, weighted precision 0.8918, recall 0.8794, F1 0.8761, and AUC-ROC 0.9534 (Table II). With TTA it reaches 87.55%, 0.8716 F1, and 0.9541 AUC, slightly lower accuracy and F1 but slightly higher AUC. The 226 of 257 correct predictions give a 95% Wilson interval of 83.39–91.37%. Agreement beyond chance is high ($\kappa = 0.8443$, $MCC = 0.8484$), but macro recall (0.8235) is well below weighted recall (0.8794), and macro F1 (0.8485) below weighted F1, because errors concentrate in the minority classes.

B. Per-Class Performance

Table III recovers one-vs-rest counts from precision, recall, and support ($TP = Rn$, $FP = TP/P - TP$); they sum to 226 correct and reproduce the weighted precision, recall, and F1. Tinea Corporis has the best F1 (0.9412) and MCC (0.9304), then Scabies (0.9027) and Contact Dermatitis (0.8774). Contact Dermatitis absorbs the most false positives (15, FPR 0.0811). Eczema has high precision (0.9815) but low recall (0.7571), missing 17 cases. Minority Atopic and Seborrheic Dermatitis have recalls of only 0.7000 and 0.5833 (FNR 0.3000 and 0.4167) but near-perfect specificity (0.9960 and 1.0000), so the model under-predicts them. Imbalance remains a major challenge despite weighted sampling and focal loss.

C. Controlled Ablation Study

On the same 257-image partition we compare (1) Swin-Tiny without graph reasoning, with Mixup/CutMix and TTA; (2) SkinDisNet-GT without TTA; (3) without Mixup/CutMix; and (4) full SkinDisNet-GT, beside the TTA variant and the two baselines (Table IV).

Removing Mixup/CutMix lowers accuracy from 87.94% to 85.60%, a 2.34 percentage-point drop (six more errors, 19.4% more error), showing its effectiveness in the pipeline. TTA does not help (87.55% and 0.8716 F1 with TTA, one more error), improving only AUC slightly. Most importantly,

the advanced-training Swin-Tiny-only configuration reaches 89.11%, surpassing SkinDisNet-GT's 87.94%. The margin is three test images, and the two Wilson intervals overlap almost entirely, so a single 257-image partition cannot separate the models. Graph reasoning alone thus shows no independent gain over the stronger Swin-Tiny baseline; rather, the full pipeline is feasible and Mixup/CutMix has a measurable effect. SkinDisNet-GT does improve on the plain ViT-B/16 and Swin-Tiny baselines by 8.17 and 3.50 points, though these baselines use a simpler training recipe. Architecture-controlled experiments are needed to isolate the contribution of graph reasoning.

D. Training Dynamics and Confusion Analysis

Over the 60-epoch run (Fig. 4a), training loss decreases rapidly, and validation accuracy and weighted F1 rise during the early and middle stages; the checkpoint is selected at the best validation weighted F1 of 0.9251 at epoch 47. The confusion matrix (Fig. 4b) shows strong diagonals for Scabies and Tinea Corporis and more evident minority-class errors for Atopic and Seborrheic Dermatitis, consistent with the per-class recalls in Table III.

E. Explainability with Grad-CAM

Grad-CAM is computed on the 196 graph-transformer node features before global mean pooling; the 14×14 activation map is resized to the input resolution and overlaid on the clinical image. For Atopic Dermatitis and Eczema (Fig. 4c), the model focuses on localized skin regions, which is qualitative evidence of attention, not clinical lesion localization.

Parameters. SkinDisNet-GT has 16,091,749 parameters versus 27,816,960 (27,519,354 in the backbone) for the Swin-Tiny ablation classifier, thanks to its compact graph-transformer head, though KNN graph construction and graph attention add operations that parameter count does not capture.

VI. LIMITATIONS AND FUTURE WORK

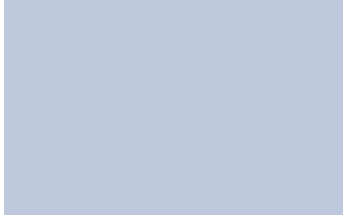
(1) Splitting is not strictly patient-level; patient-grouped stratified splits with reliable metadata-to-image matching are needed. (2) Swin-Tiny-only outperforms the current model, so identical training with and without the graph module must isolate its gain. (3) A single cohort and test partition call for external validation, patient-level cross-validation, calibration, and subgroup robustness analysis. (4) Grad-CAM is not clinical validation of lesion localization and needs systematic evaluation and expert assessment. Graph-size sensitivity, structural encoding ablations, and uncertainty estimation are further directions.

VII. CONCLUSION

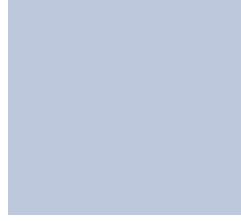
SkinDisNet-GT turns a Swin-Tiny 14×14 feature grid into a 196-node, nine-nearest-neighbor graph processed by structural encoding, graph aggregation, multi-head attention, and three graph-transformer blocks, trained with imbalance-aware sampling, focal loss, and Mixup/CutMix. It achieved 87.94% accuracy, 0.8761 weighted F1, 0.9534 weighted AUC-ROC, and 0.6060 test loss ($\kappa = 0.8443$, $MCC = 0.8484$)

TABLE III
SKINDISNET-GT PER-CLASS ONE-VS-REST TEST RESULTS. TP – TN ARE DERIVED FROM THE REPORTED PRECISION, RECALL, AND SUPPORT; SENS. = RECALL, SPEC. = SPECIFICITY, BA = BALANCED ACCURACY, J = JACCARD INDEX

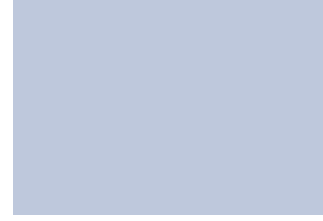
Class	TP	FP	FN	TN	Prec.	Sens.	Spec.	NPV	F_1	F_2	J	BA	MCC	FPR	FNR	Supp.
Atopic Derm.	7	1	3	246	0.8750	0.7000	0.9960	0.9880	0.7778	0.7292	0.6364	0.8480	0.7750	0.0040	0.3000	10
Contact Derm.	68	15	4	170	0.8193	0.9444	0.9189	0.9770	0.8774	0.9164	0.7816	0.9317	0.8291	0.0811	0.0556	72
Eczema	53	1	17	186	0.9815	0.7571	0.9947	0.9163	0.8548	0.7934	0.7465	0.8759	0.8215	0.0053	0.2429	70
Scabies	51	10	1	195	0.8361	0.9808	0.9512	0.9949	0.9027	0.9480	0.8226	0.9660	0.8800	0.0488	0.0192	52
Seborrheic Derm.	7	0	5	245	1.0000	0.5833	1.0000	0.9800	0.7368	0.6364	0.5833	0.7917	0.7561	0.0000	0.4167	12
Tinea Corporis	40	4	1	212	0.9091	0.9756	0.9815	0.9953	0.9412	0.9615	0.8889	0.9785	0.9304	0.0185	0.0244	41
Total / Macro	226	31	31	1254	0.9035	0.8235	0.9737	0.9752	0.8485	0.8308	0.7432	0.8986	0.8320	0.0263	0.1765	257
Weighted	–	–	–	–	0.8918	0.8794	0.9628	0.9676	0.8761	0.8761	0.7825	0.9211	0.8480	0.0372	0.1206	257



(a) Loss, accuracy, weighted F1, AUC over 60 epochs



(b) Confusion matrix (counts, row %)



(c) Grad-CAM: Atopic Derm. and Eczema

Fig. 4. SkinDisNet-GT training dynamics, confusion analysis on the test set, and Grad-CAM explanations.

TABLE IV
BASELINE AND CONTROLLED ABLATION RESULTS ON THE 257-IMAGE TEST SET. Δ VALUES ARE RELATIVE TO SKINDISNET-GT (NO TTA); REL. ERR. IS THE ERROR-RATE RATIO e/e_{GT} (BELOW 1 MEANS FEWER ERRORS)

Configuration	Group	Correct	Acc.	95% Wilson CI	Error	Δ Acc. (pp)	Rel. Err.	W-F1	Δ F1	AUC
ViT-B/16	Baseline	205	79.77%	[74.44, 84.23]	20.23%	–8.17	1.677	0.7948	–0.0813	–
Swin-Tiny	Baseline	217	84.44%	[79.50, 88.36]	15.56%	–3.50	1.290	0.8406	–0.0355	–
Swin-Tiny only (adv. training)	Ablation	229	89.11%	[84.71, 92.36]	10.89%	+1.17	0.903	0.8896	+0.0135	0.9812
SkinDisNet-GT, no TTA	Proposed	226	87.94%	[83.39, 91.37]	12.06%	0.00	1.000	0.8761	0.0000	0.9534
SkinDisNet-GT, with TTA	Ablation	225	87.55%	[82.95, 91.04]	12.45%	–0.39	1.032	0.8716	–0.0045	0.9541
SkinDisNet-GT, no Mixup/CutMix	Ablation	220	85.60%	[80.78, 89.37]	14.40%	–2.34	1.194	0.8519	–0.0242	0.9540
SkinDisNet-GT, full	Ablation	226	87.94%	[83.39, 91.37]	12.06%	0.00	1.000	0.8761	0.0000	0.9534

on 257 test images. Removing Mixup/CutMix reduced accuracy to 85.60%, while the advanced-training Swin-Tiny-only model reached 89.11% within overlapping confidence intervals, so the results demonstrate feasibility but not an independent gain from graph reasoning. Patient-level evaluation and architecture-controlled experiments are needed to establish it.

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